

BLUESTOCK
MUTUAL FUND ANALYTICS
CAPSTONE PROJECT

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Executive Summary:

The Bluestock Mutual Fund Analytics Capstone Project was developed to analyze mutual fund performance, investor behavior, and market trends using data analytics and business intelligence tools. The project involved data collection, cleaning, transformation, storage in SQLite, exploratory data analysis, performance analytics, and dashboard development.

Advanced analytics such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Investor Cohort Analysis, SIP Continuity Analysis, and a Fund Recommendation System were implemented to generate meaningful insights. An interactive Power BI dashboard was created to visualize key metrics and trends.

The project successfully demonstrates the application of Python, SQL, SQLite, Jupyter Notebook, and Power BI in building an end-to-end mutual fund analytics solution that supports data-driven investment decision-making.

Introduction:

Problem Statement:

Investors often face challenges in:

- Evaluating fund performance objectively.
- Understanding risk-adjusted returns.
- Tracking investor behavior trends.
- Comparing funds against benchmarks.
- Monitoring SIP continuity and investment discipline.

The absence of a unified analytics platform makes these tasks difficult. Therefore, this project develops a centralized mutual fund analytics solution.

Objectives:

The project objectives include:

1. Build an ETL pipeline for mutual fund datasets.
2. Store cleaned data in SQLite.
3. Perform exploratory data analysis.
4. Compute key performance metrics.
5. Build an interactive dashboard.
6. Develop advanced analytics models.
7. Generate actionable business insights.

Data Sources:

Eight datasets were used in this project.

Dataset 1: Fund Master

Contains information about mutual fund schemes, fund houses, categories, and plan types.

Key fields:

- amfi_code
- scheme_name
- fund_house
- category
- plan

Dataset 2: NAV History

Contains historical Net Asset Value information.

Key fields:

- amfi_code
- nav_date
- nav

Dataset 3: AUM by Fund House

Contains Assets Under Management information for fund houses.

Dataset 4: Category Inflows

Contains monthly inflows and outflows across categories.

Dataset 5: Industry Portfolio Holdings

Contains sector allocation and portfolio composition.

Dataset 6: Benchmark Indices

Contains benchmark index performance data.

Dataset 7: Scheme Performance

Contains return metrics and risk indicators.

Fields include:

- Return 1Y
- Return 3Y
- Return 5Y
- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio

Dataset 8: Investor Transactions

Contains transaction-level investor data.

Fields include:

- investor_id
- transaction_date
- transaction_type
- amount_inr
- state
- city
- age_group
- gender
- payment_mode

ETL Pipeline Design

3.1 Data Ingestion:

Raw CSV files were collected and loaded using Python Pandas. Individual scripts were developed for each dataset.

3.2 Data Cleaning:

The following cleaning operations were performed:

- Missing value handling.
- Duplicate record removal.
- Standardization of column names.
- Date format conversion.
- Numeric datatype correction.

3.3 Data Validation:

Validation scripts checked:

- Missing values.
- Invalid AMFI codes.
- Negative transaction amounts.
- Duplicate records.
-

3.4 SQLite Database Implementation:

A relational SQLite database was created.

Tables:

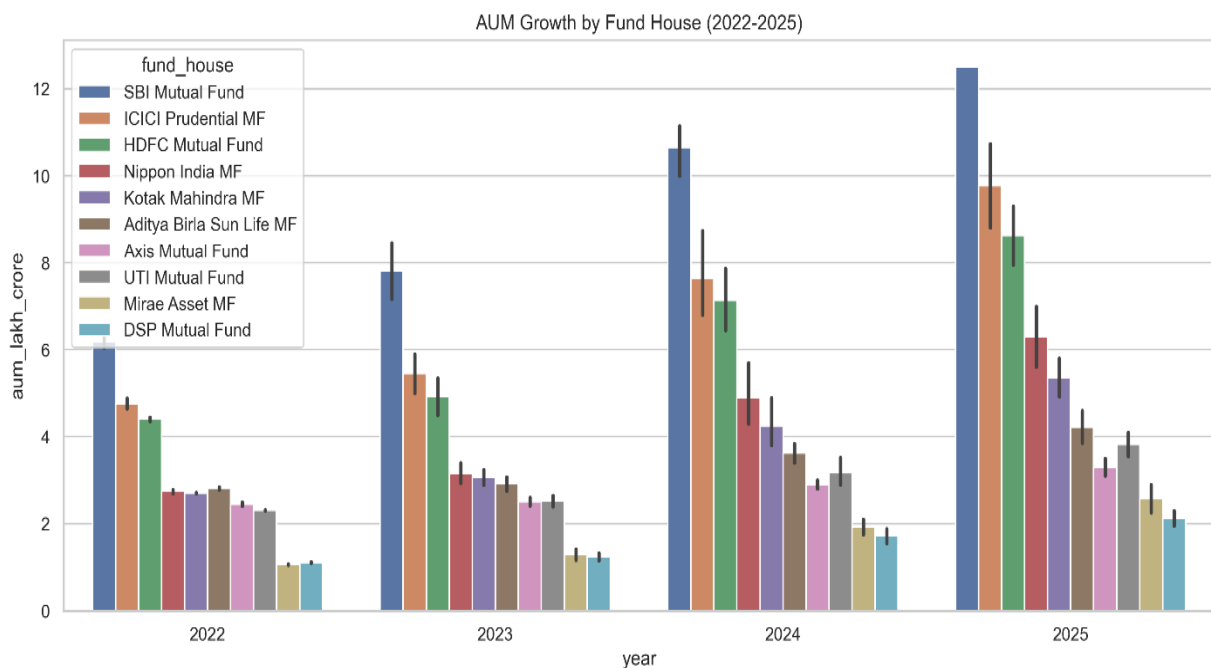
- fact_nav
- fact_transactions
- fact_performance

The database enables efficient querying and analytics.

4. Exploratory Data Analysis

4.1 AUM Growth Analysis

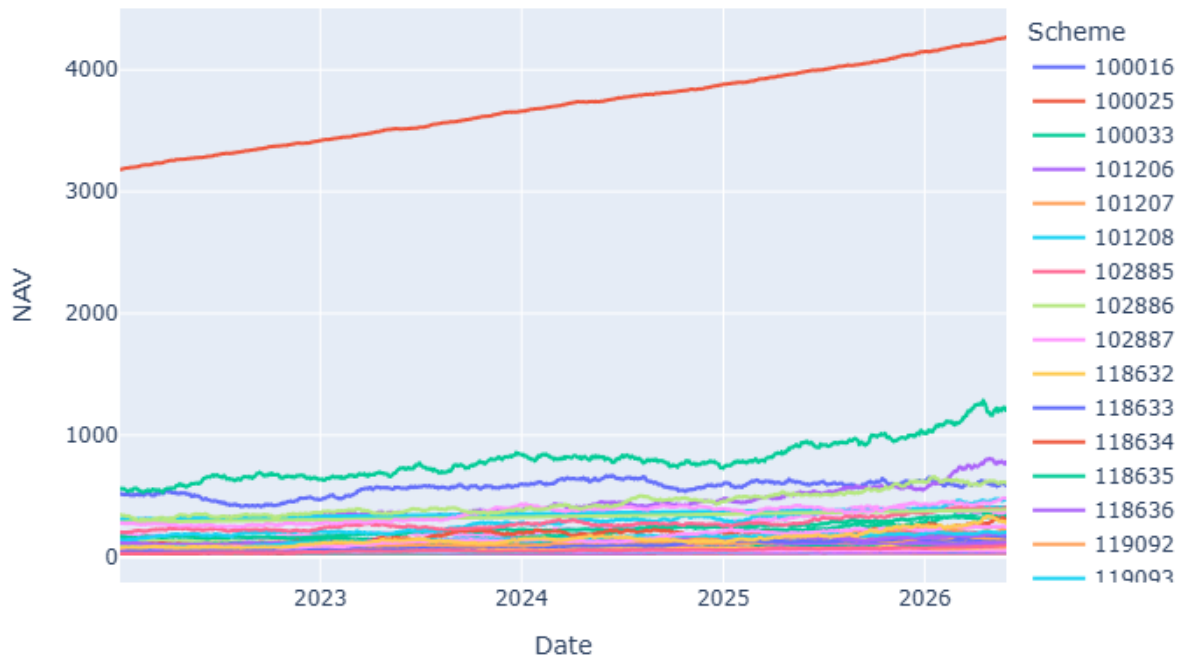
The AUM analysis revealed significant growth among large-cap and diversified equity funds. Large fund houses consistently attracted higher investments due to investor confidence and brand reputation.



4.2 NAV Trend Analysis

NAV trends indicated long-term growth in major equity funds. Market downturns were visible during correction periods, followed by recovery phases.

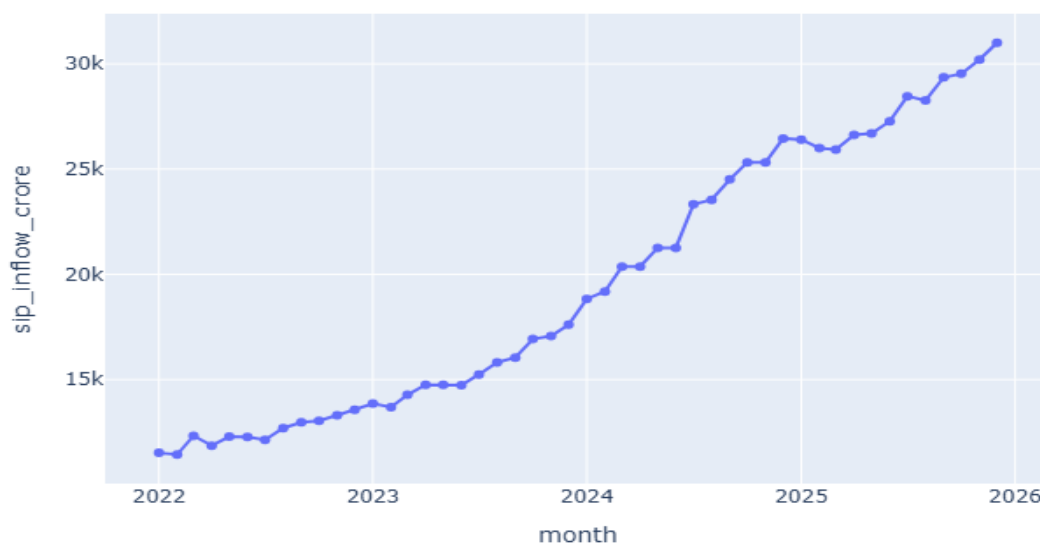
Daily NAV Trend for All Mutual Fund Schemes (2022–2026)



4.3 SIP Inflow Analysis

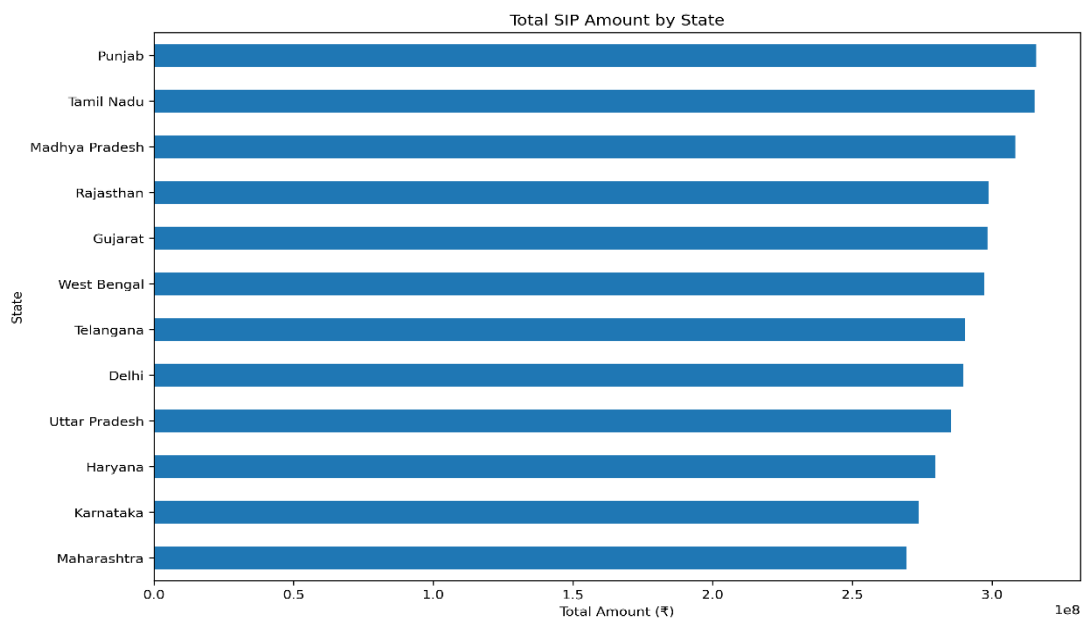
SIP contributions showed steady growth over time, highlighting increasing investor participation in systematic investing.

Monthly SIP Inflow Trend (Jan 2022 – Dec 2025)



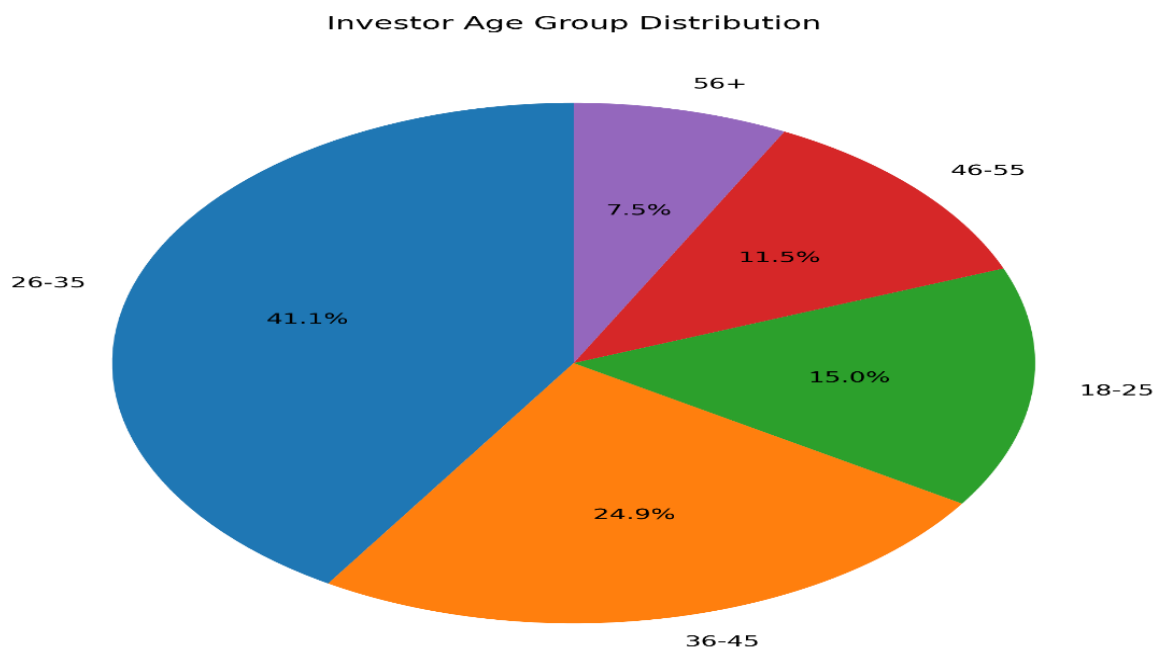
4.4 Geographic Distribution

Metro cities such as Mumbai, Hyderabad, Bengaluru, and Delhi contributed the highest transaction volumes.



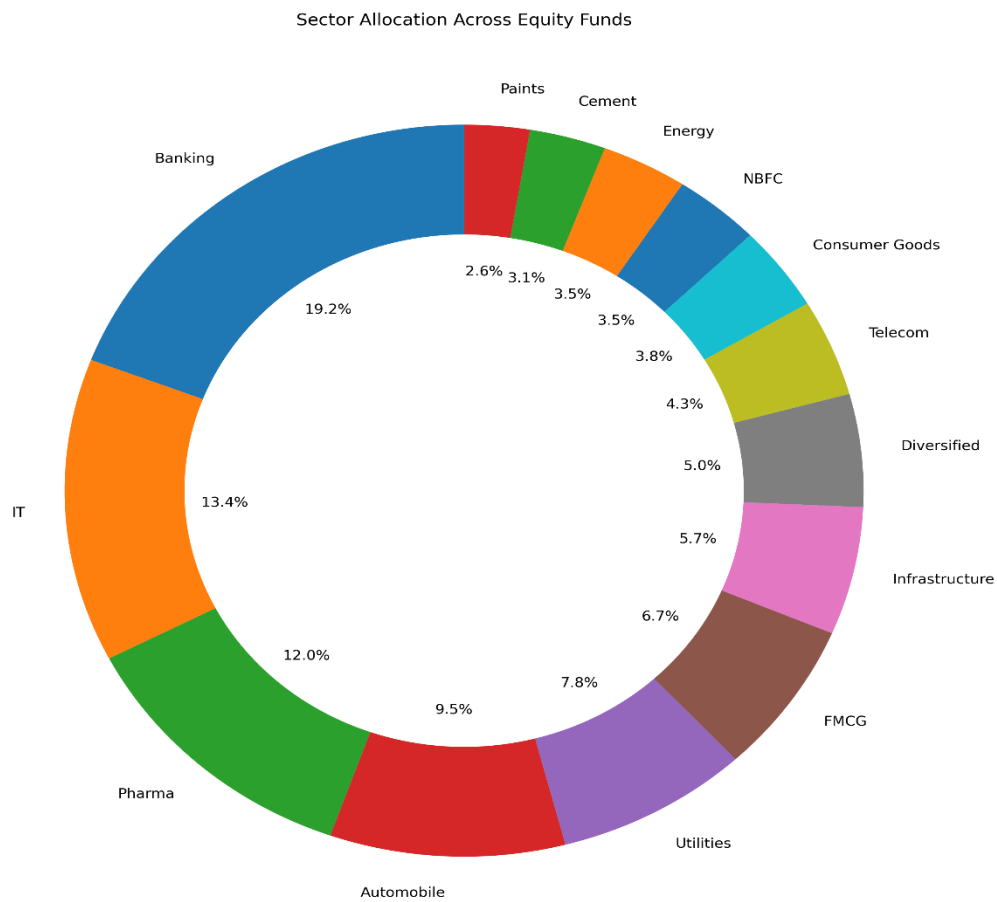
4.5 Investor Demographics

The age group 26–35 represented the largest investor segment. Male investors constituted a larger share of transactions compared to female investors.



4.6 Sector Allocation Analysis

Technology, Banking, and Financial Services sectors accounted for major portfolio allocations across equity schemes.

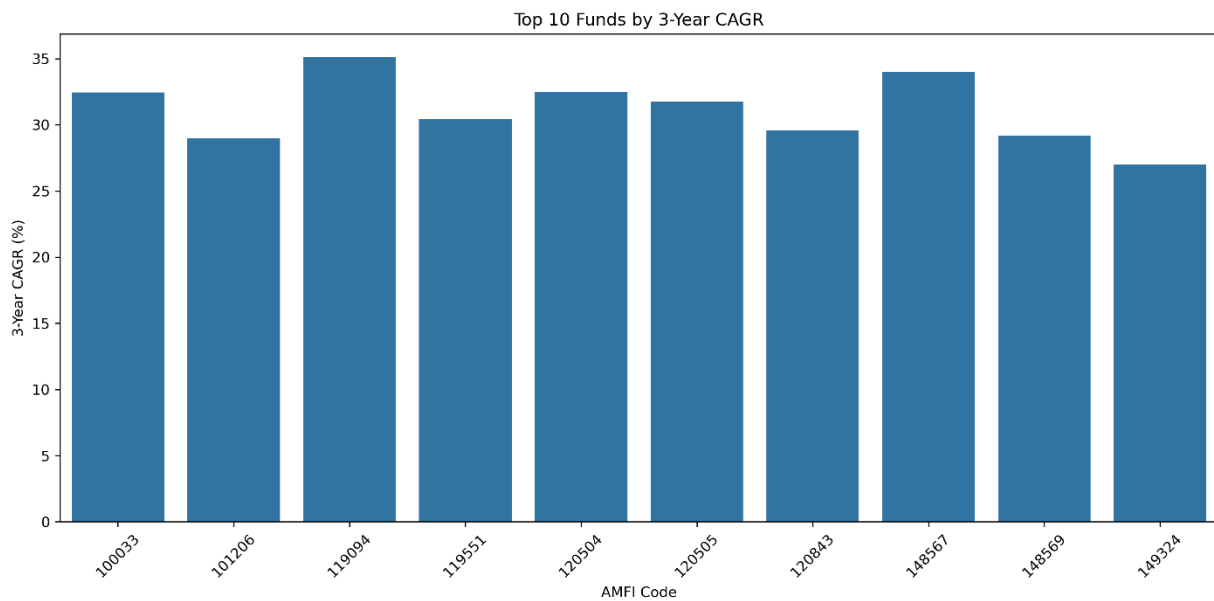


5. Performance Analytics

5.1 CAGR Analysis

Compound Annual Growth Rate was used to evaluate long-term growth performance.

Results showed that equity-oriented funds generated superior long-term returns compared to conservative categories.

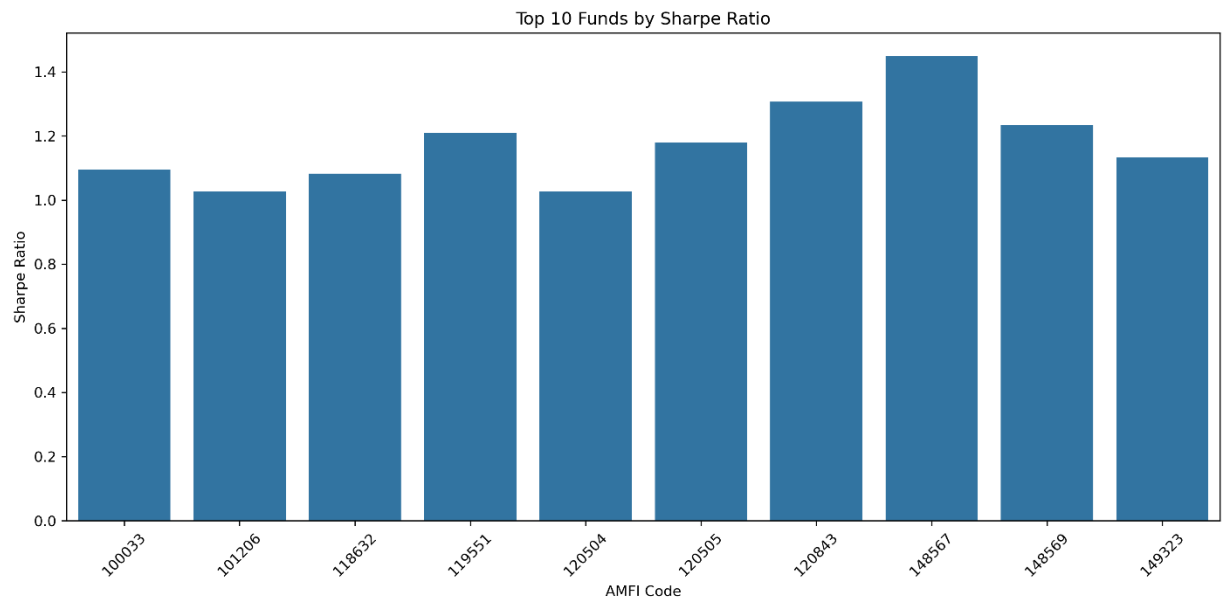


5.2 Sharpe Ratio Analysis

Sharpe Ratio measures risk-adjusted returns.

Higher Sharpe Ratio values indicate better reward per unit of risk.

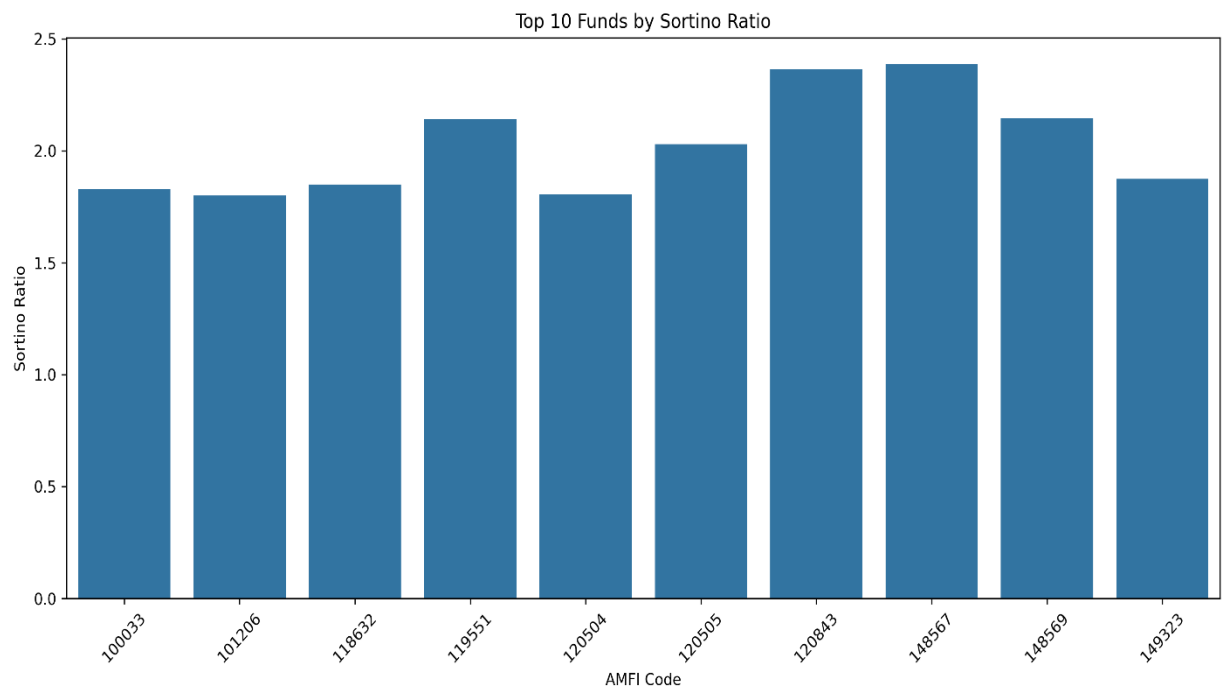
Top-performing funds demonstrated strong risk-adjusted performance.



5.3 Sortino Ratio Analysis

Sortino Ratio focuses only on downside risk.

Several funds exhibited stronger Sortino Ratios compared to Sharpe Ratios, indicating resilience against negative returns.



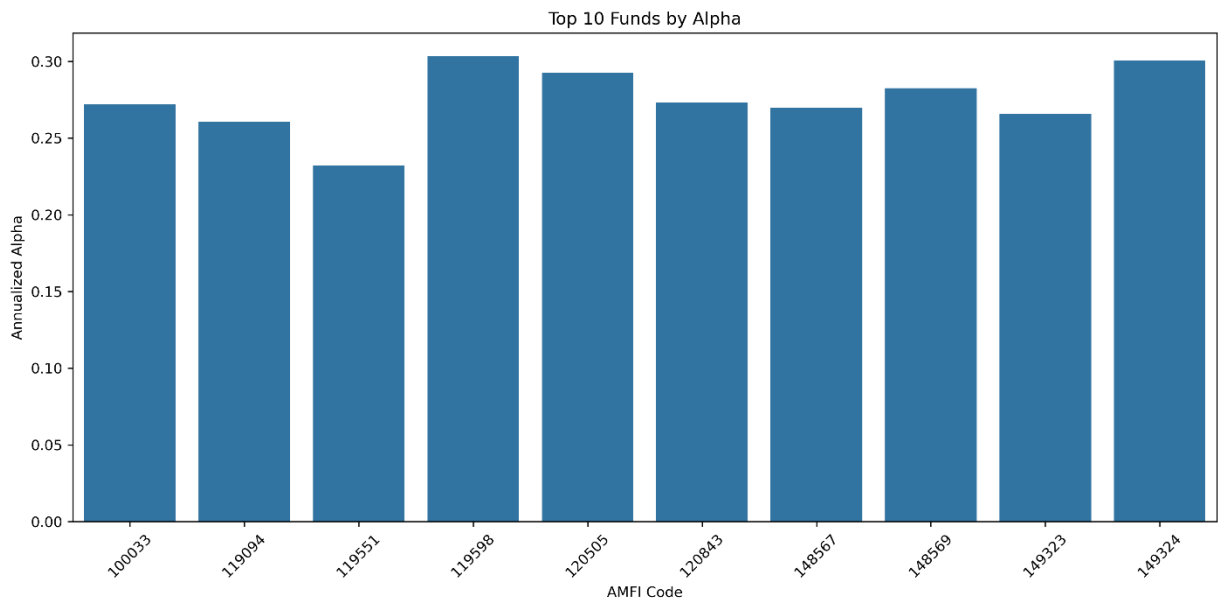
5.4 Alpha and Beta Analysis

Alpha measured excess returns generated over benchmarks.

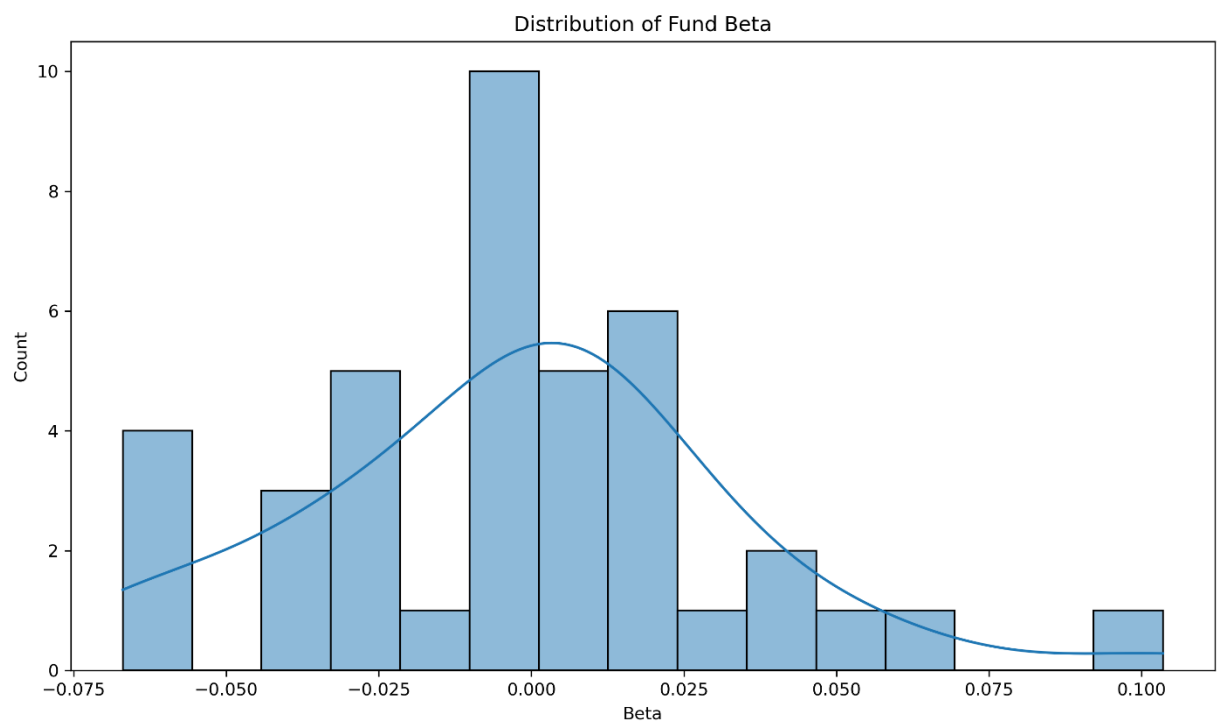
Beta measured sensitivity to market movements.

The analysis identified both defensive and aggressive funds.

Alpha Ranking:

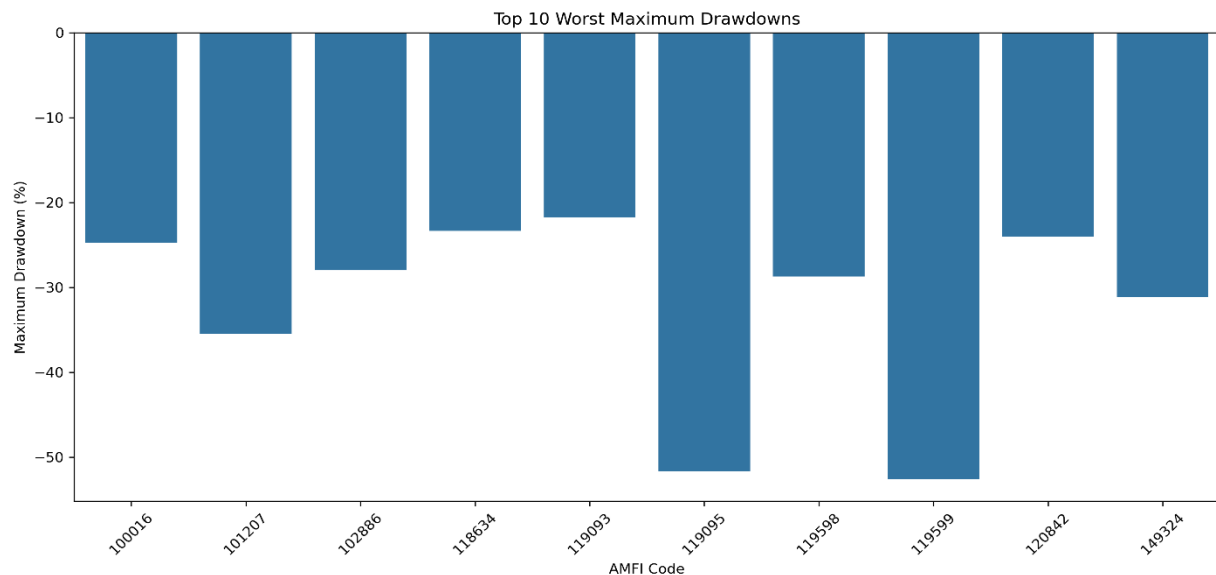


Beta Distribution:



5.5 Maximum Drawdown

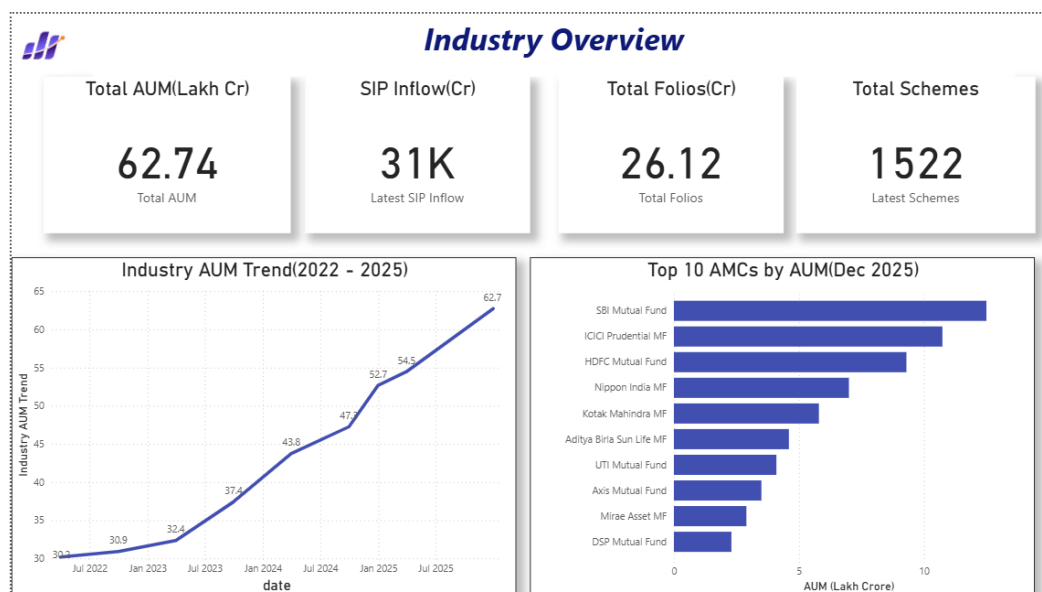
Maximum Drawdown analysis highlighted downside exposure during adverse market periods.



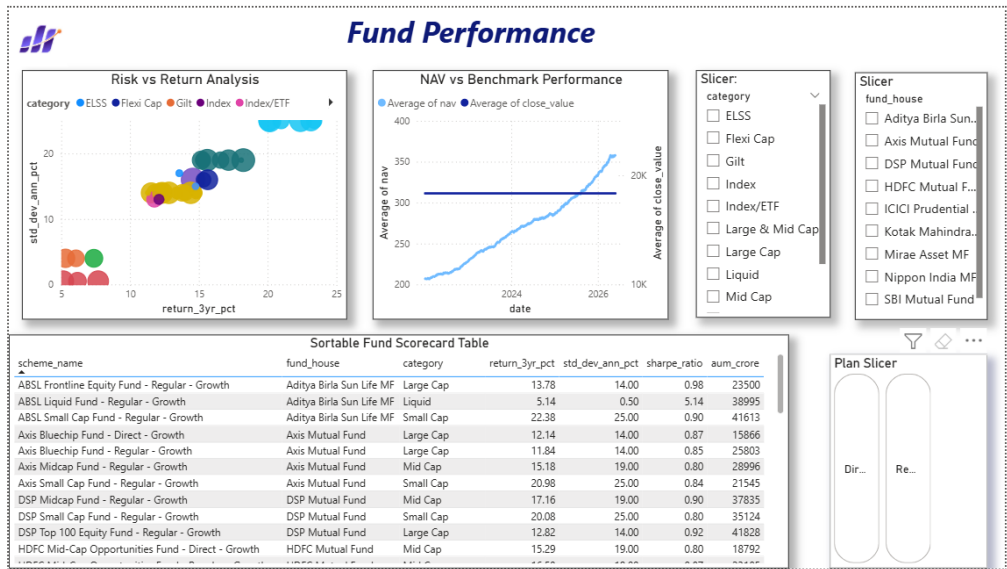
6. Interactive Dashboard

A four-page Power BI dashboard was developed.

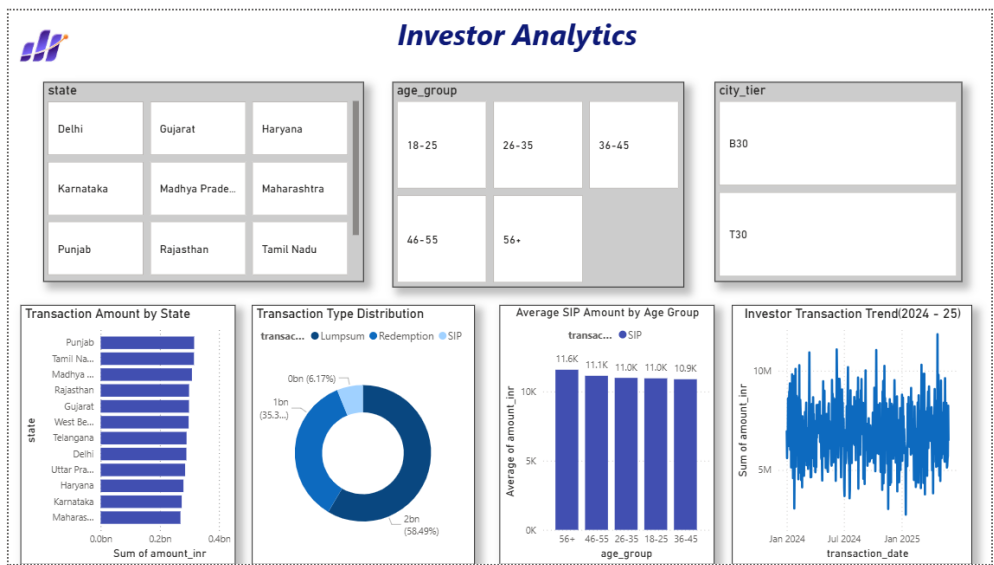
Page 1: Industry Overview



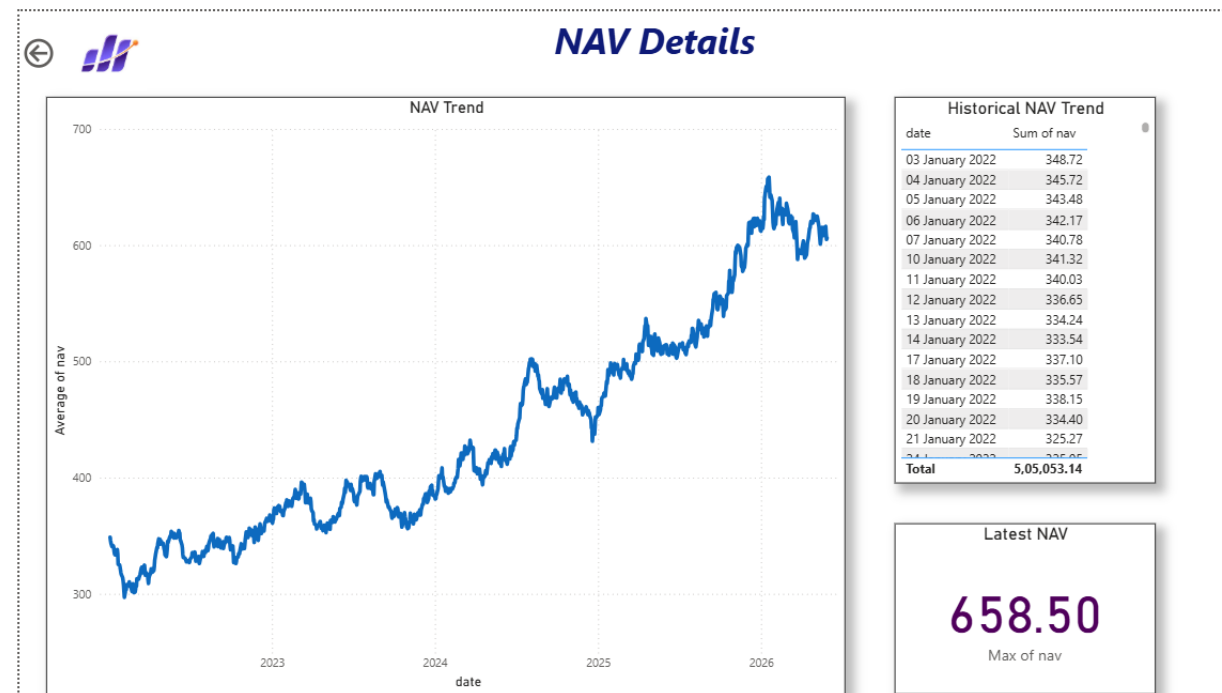
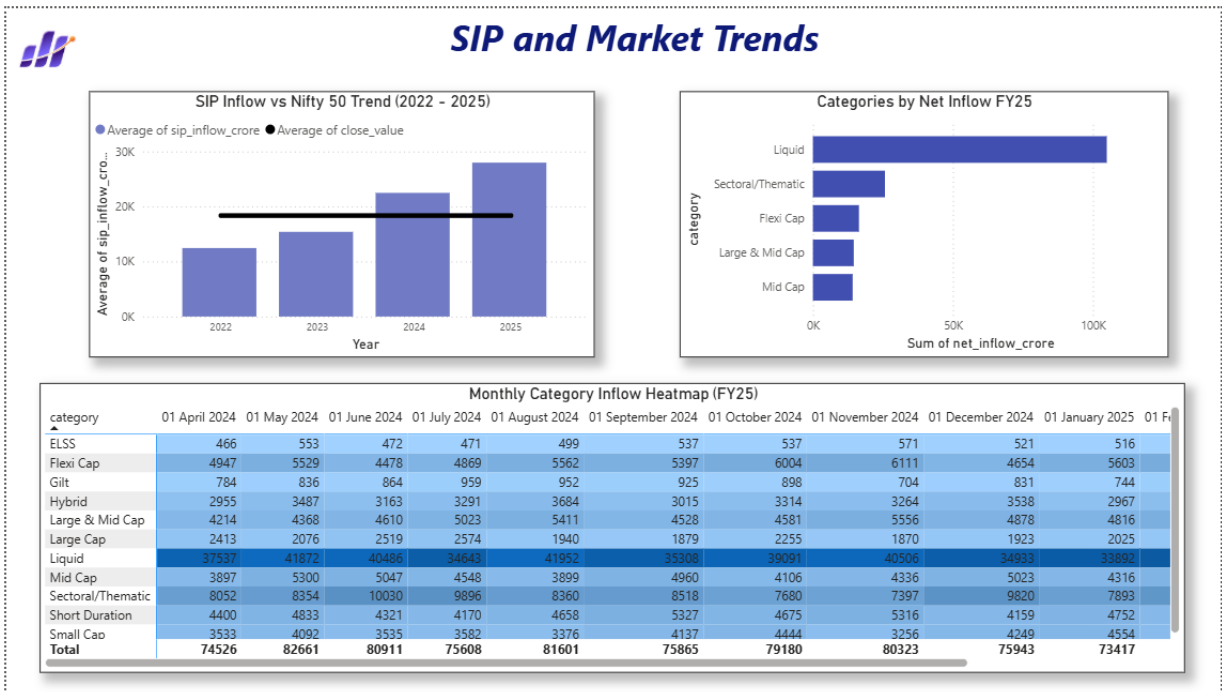
Page 2: Fund Performance



Page 3: Investor Analysis



Page 4: SIP Market Trends



7. Advanced Analytics

7.1 Value at Risk (VaR)

VaR estimates the maximum expected loss at a specified confidence level.

The analysis identified funds with higher downside risk exposure.

7.2 Conditional Value at Risk (CVaR)

CVaR measured expected losses beyond the VaR threshold.

CVaR provided a deeper understanding of tail risk.

7.3 Investor Cohort Analysis

Investors were grouped based on their first transaction year.

The cohort analysis revealed investor retention patterns and investment growth trends.

7.4 SIP Continuity Analysis

Average transaction gaps were calculated.

Investors with gaps exceeding 35 days were classified as at-risk investors.

7.5 Fund Recommendation System

A rule-based recommendation engine was developed.

Risk Categories:

- Low Risk
- Moderate Risk
- High Risk

Recommendations were generated using Sharpe Ratio rankings.

7.6 Sector HHI Concentration Analysis

Herfindahl-Hirschman Index (HHI) was calculated to measure portfolio concentration.

Higher HHI values indicated concentrated portfolios, while lower values indicated diversification.

Key Findings

1. Equity funds generated superior long-term returns.
2. SIP investments showed steady growth.
3. Investors aged 26–35 dominate transaction activity.
4. Metro cities contribute most investments.
5. Several funds achieved strong Sharpe Ratios.
6. HHI analysis revealed concentration risks in selected funds.
7. Cohort analysis indicated improving investor retention.

Conclusion

The Mutual Fund Analytics Capstone Project successfully developed an end-to-end analytics platform integrating ETL processes, database management, performance analytics, dashboard visualization, and advanced analytics. The project demonstrates how data-driven techniques can be applied to evaluate mutual fund performance, investor behavior, and risk characteristics, ultimately supporting more informed investment decisions.

